

MODEL QUESTION PAPER SET I

Hospital Systems

Part A

Answer all questions in one word or one sentence.

- 1 What is the flaw in the argument that “X ray room dimension in India should be as per FDA guidelines”
2. Who is the top official of a Standard 300 bed hospitals?
3. What is the standard colour coding of oxygen and air for Medical gas pipeline system in hospital?
4. What is the definition for macro shock?
5. Identify and name two types of supplies that can be sterilized using an autoclave
6. List any two physiological parameters adaptable to telemetric system.
7. A system set up where the doctors can remotely monitor various vital signs of ambulatory patients is called.....
8. Write one significance for calibration of medical equipment?
9. What is the abbreviation of NABH ?

1*9=9 marks

PART B

II. Answer any eight questions from the following, Each question carries 3marks.

- 1.Differentiate between Clinical Engineer and Hospital Engineer?
- 2.List 3 duties of Biomedical Engineer related to equipment Maintenance ?
- 3.In the planning stage of hospital, list 3 instructions to be given to architect regarding the layout of X ray room?
- 4.What are the three types of power sources in a hospital?
- 5.Explain the physiological effects of electric current?
- 6.Describe the significance of steam sterilizer in CSSD department?
- 7.List 3 difference between Amplitude modulation and Pulse modulation?

8. Identify the system components of Biotelemetry system with suitable block diagram?

9. Explain in house contract management ?

10. Explain the significance of ISO in hospital accreditation?

3* 8=24 marks

PART C

Answer ALL questions. Each question carries 7 marks

III Explain hospital organization chart with diagram?

OR

IV Design& Explain ICU in a hospital with suitable layout?

V List the duties and responsibilities of a Biomedical Engineer?

OR

VI Design & Explain Emergency department of a hospital with suitable layout?

VII Draw and explain the schematic of power distribution system in a hospital?

OR

VIII Explain any 3 basic electric safety measures adopted in a hospital?

IX Describe with schematic the MGPS for Centralized gas supply?

OR

X Explain any 4 sterilization technique and the materials sterilized by that technique?

XI Describe single channel telemetry with block diagram?

OR

XII ExplainASK concept and wave form in detail?

XIII Explain Joint commission International and its significance

OR

XIV Describe the types of equipment maintenance in detail.

6*7=42 marks

MODEL QUESTION PAPER SET II

Hospital Systems

Part A

Answer all questions in one word or one sentence.

- 1 What is the flaw in the argument that “No diagnostic X-ray equipment shall be operated for patient diagnosis unless Licence for operation is obtained from the Competent Authority.”
2. Who is responsible for recruitment in a standard 300 bedded hospital?
3. What is the standard colour coding of vacuum and nitrous oxide for Medical gas pipeline system in hospital?
4. What cause microshock?
5. Identify and name two types of supplies that can be sterilized using an ETO
6. Define bio telemetric system.
7. What is the significance of transmitter in a communication system?
8. Write one advantage of calibrating the medical equipment?
9. List any 2 reasons why we have to do maintenance of equipment ?

1*9=9marks

PART B

II. Answer any eight questions from the following, Each question carries 3marks.

1. How does Biomedical Engineer and Bio Engineer?
2. List any 5 design considerations while planning operation theatre ?
3. Explain any 2 types of health care organizations?
4. Draw and explain sterile supply cycle?
5. Explain voltage stabilizer and its application in a hospital environment?
6. With a neat diagram explain double chamber autoclave used in CSSD department?
7. Differentiate ASK from FSK?

8. Explain 3 physiological parameter in biotelemetry system?
9. Explain why accreditation is important in hospitals ?
10. List any 4 differences between preventive maintenance & break down maintenance ?

3* 8=24 marks

PART C

Answer ALL questions. Each question carries 7 marks

III Explain functional planning of a hospital?

OR

IV Design& Explain X ray room in a hospital with suitable layout?

V List the duties and responsibilities of a Biomedical Engineer?

OR

VI Design & Explain patient ward of a hospital with suitable layout?

VII Explain different types of power sources used in hospital?

OR

VIII Explain any 3 basic electric safety measures adopted in a hospital?

IX Describe with schematic Centralized vacuum supply in hospitals?

OR

X Explain any one type of sterilizer used in hospitals with suitable diagram?

XI Describe double channel telemetry with block diagram?

OR

XII Explain PSK concept and wave form in detail?

XIII Explain ISO certification process in hospitals

OR

XIV Describe the general procedure for testing and calibration of medical equipment.

6*7=42marks